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Quartz crystal resonator unit, electronic component, and electronic device

Abstract

A quartz crystal resonator unit that includes: a quartz crystal resonator including an excitation electrode and a connection electrode electrically connected to the excitation electrode; a base member including an electrode pad; an electrically conductive holding member that connects the connection electrode and the electrode pad to each other; and a lid member attached to the base member so as to form an internal space between the lid member and the base member in which the quartz crystal resonator is accommodated. The electrically conductive holding member is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main component and (2) an epoxy compound having two or more epoxy groups.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of International application No. PCT/JP2020/023184, filed Jun. 12, 2020, which claims priority to Japanese Patent Application No. 2019-138921, filed Jul. 29, 2019, the entire contents of each of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to a quartz crystal resonator unit, an electronic component, and an electronic device.

BACKGROUND OF THE INVENTION

(2) Piezoelectric resonator units are being used as timing devices, sensors, oscillators, etc. in various electronic devices such as mobile communication terminals, communication base stations, and home appliances. For example, a piezoelectric resonator unit includes a piezoelectric resonator that has a mechanical vibration portion that converts electric vibration into mechanical vibration by using a piezoelectric effect, a holder that accommodates the piezoelectric resonator, and an electrically conductive holding member that connects the piezoelectric resonator and the holder to each other. The electrically conductive holding member is, for example, a cured product of an electrically conductive adhesive containing an epoxy resin as a main component.

(3) Patent Document 1 discloses an electrically conductive adhesive that contains an epoxy resin including 5 to 60 wt % of glycidyl siloxane, a curing agent, and an electrically conductive material.

(4) Patent Document 2 discloses an electrically conductive adhesive including an epoxy resin in which a glycidyl reactive diluent accounts for 20 to 70 wt %, a phenol resin curing agent in which an alkyl resol phenol resin and/or an alkyl novolac phenol resin accounts for 50 wt % or more, and electrically conductive particles.

(5) Patent Document 3 discloses a quartz crystal resonator unit in which a metal electrode provided on a quartz crystal element and one end of an extended conductor of an envelope are bonded and fixed to each other with an electrically conductive adhesive containing, as essential components, an organic resin having two or more carboxyl groups in its molecular side chain or molecule end, an epoxy compound having an alicyclic epoxy and a glycidyl group in one molecule, and electrically conductive powder. Patent Document 1: Japanese Unexamined Patent Application Publication No. 59-172571 Patent Document 2: Japanese Unexamined Patent Application Publication No. 9-157613 Patent Document 3: Japanese Unexamined Patent Application Publication No. 2-180975

SUMMARY OF THE INVENTION

(6) In recent years, studies have been made to improve frequency-temperature characteristics by using, as an electrically conductive holding member that connects a quartz crystal resonator and a holder to each other, a cured product of a silicone-based electrically conductive adhesive containing a silicone resin as a main component. However, when a cured product of a resin-based adhesive is used as a joining member that connects a base member and a lid member, the use of a cured product of a silicone-based electrically conductive adhesive as an electrically conductive holding member has caused a problem of deterioration of resistance in a high-humidity environment.

(7) The present invention has been made in view of these circumstances, and an object thereof is to provide a quartz crystal resonator unit with improved reliability, an electronic component, and an electronic device.

(8) A quartz crystal resonator unit according to one aspect of the present invention includes: a quartz crystal resonator including an excitation electrode and a connection electrode electrically connected to the excitation electrode; a base member including an electrode pad; an electrically conductive holding member that connects the connection electrode and the electrode pad to each

other; and a lid member attached to the base member so as to form an internal space between the lid member and the base member in which the quartz crystal resonator is accommodated. The electrically conductive holding member is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main component and (2) an epoxy compound having two or more epoxy groups.

(9) An electronic device according to one aspect of the present invention includes an electronic component, a substrate including a metal layer, and an electrically conductive adhesive portion that connects the electronic component and the metal layer of the substrate to each other. The electrically conductive adhesive portion is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main component and (2) an epoxy compound having two or more epoxy groups.

(10) An electronic component according to one aspect of the present invention includes a first adherend portion having electrical conductivity, a second adherend portion having electrical conductivity, and an electrically conductive adhesive portion that connects the first adherend portion and the second adherend portion to each other. The electrically conductive adhesive portion is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main component and (2) an epoxy compound having two or more epoxy groups.

(11) According to the present invention, a quartz crystal resonator unit with improved reliability, an electronic component, and an electronic device can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an exploded perspective view schematically illustrating the configuration of a quartz crystal resonator unit according to a first embodiment.

(2) FIG. 2 is a sectional view schematically illustrating the configuration of the quartz crystal resonator unit according to the first embodiment.

(3) FIG. 3 shows a structural formula of an epoxy compound added to an electrically conductive adhesive.

(4) FIG. 4 is a plan view schematically illustrating the structure of an evaluation substrate used in a moisture resistance test.

(5) FIG. 5 is a sectional view schematically illustrating the structure of the evaluation substrate used in the moisture resistance test.

(6) FIG. 6 is a graph schematically showing the evaluation results of an example.

(7) FIG. 7 is a graph schematically showing the evaluation results of a comparative example.

(8) FIG. 8 is a sectional view schematically illustrating the configuration of an electronic device according to a second embodiment.

(9) FIG. 9 is graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 0 wt %.

(10) FIG. 10 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 0.1 wt %.

(11) FIG. 11 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 0.5 wt %.

(12) FIG. 12 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 1 wt %.

(13) FIG. 13 is graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 5 wt %.

(14) FIG. 14 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 10 wt %.

- (15) FIG. **15** shows a structural formula of an epoxy compound added in a first modified example.
- (16) FIG. **16** is a graph showing the results of an evaluation of moisture resistance in the first modified example.
- (17) FIG. **17** shows a structural formula of an epoxy compound added in a second modified example.
- (18) FIG. **18** is a graph showing the results of an evaluation of moisture resistance in the second modified example.
- (19) FIG. **19** shows a structural formula of an epoxy compound added in a third modified example.
- (20) FIG. **20** is a graph showing the results of an evaluation of moisture resistance in the third modified example.
- (21) FIG. **21** shows a structural formula of an epoxy compound added in a fourth modified example.
- (22) FIG. **22** is a graph showing the results of an evaluation of moisture resistance in the fourth modified example.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(23) Hereinafter, embodiments of the present invention will be described with reference to the drawings. The drawings of the embodiments are given by way of illustration. The dimensions and shapes of elements are schematic, and the embodiments should not be construed to limit the technical scope of the present invention.

First Embodiment

(24) The configuration of a quartz crystal resonator unit **1** according to a first embodiment of the present invention will be described with reference to FIG. **1** to FIG. **3**. FIG. **1** is an exploded perspective view schematically illustrating the configuration of the quartz crystal resonator unit according to the first embodiment. FIG. **2** is a sectional view schematically illustrating the configuration of the quartz crystal resonator unit according to the first embodiment. FIG. **3** shows a structural formula of an epoxy compound added to an electrically conductive adhesive. The sectional view illustrated in FIG. **2** shows a section of the quartz crystal resonator unit **1** taken along line II-II of FIG. **1**.

(25) In each figure, an orthogonal coordinate system with an X-axis, a Y'-axis, and a Z'-axis may be provided for convenience in order to clarify the relationship between the figures and help understand the positional relationship of members. The X-axis, the Y'-axis, and the Z'-axis in one figure respectively correspond to the X-axis, the Y'-axis, and the Z'-axis in the other figure. The X-axis, the Y'-axis, and the Z'-axis correspond to the crystallographic axes of a quartz crystal element **11**, which will be described later. The X-axis corresponds to the electrical axis (polar axis), a Y-axis to the mechanical axis, and a Z-axis to the optical axis. The Y'-axis and the Z'-axis are axes obtained by rotating the Y-axis and the Z-axis, respectively, about the X-axis by 35 degree 15 minutes $\pm$ 1 minute 30 seconds in a direction from the Y-axis toward the Z-axis.

(26) In the following description, a direction parallel to the X-axis is referred to as an "X-axis direction", a direction parallel to the Y'-axis as a "Y'-axis direction", and a direction parallel to the Z'-axis as a "Z'-axis direction". Directions of arrows of the X-axis, the Y'-axis, and the Z'-axis are referred to as "+(plus)", and directions opposite to the arrows as "- (minus)". Although the description will be made with the +Y'-axis direction being an upward direction and the -Y'-axis direction being a downward direction for convenience, the vertical orientation of the quartz crystal resonator unit **1** is not limited. For example, in the following description, the +Y' side and the -Y' side of a quartz crystal resonator **10** are respectively referred to as an upper surface **11A** and a lower surface **11B**, but the quartz crystal resonator unit **1** may be disposed such that the upper surface **11A** is located vertically below the lower surface **11B**.

(27) The quartz crystal resonator unit **1** includes a quartz crystal resonator **10**, a base member **30**, a lid member **40**, and a joining member **50**. The quartz crystal resonator **10** is provided between the base member **30** and the lid member **40**. The base member **30** and the lid member **40** constitute a

holder for accommodating the quartz crystal resonator **10**. In the example illustrated in FIG. 1 and FIG. 2, the base member **30** is shaped like a flat plate, and the lid member **40** defines a bottomed hollow space for accommodating the quartz crystal resonator **10** on the base member **30** side. The quartz crystal resonator **10** is mounted on the base member **30**. The base member **30** and the lid member **40** may have any other shapes as long as at least a portion of the quartz crystal resonator **10** that is to be excited is accommodated in the holder. In addition, the quartz crystal resonator **10** may be held in any other manner. For example, the base member **30** may define a bottomed hollow space for accommodating the quartz crystal resonator **10** on the lid member **40** side. The base member **30** and the lid member **40** may sandwich a peripheral portion of a portion of the quartz crystal resonator **10** that is to be excited.

(28) First, the quartz crystal resonator **10** will be described.

(29) The quartz crystal resonator **10** is a device that converts electrical energy and mechanical energy into each other by vibrating quartz crystal using a piezoelectric effect. The quartz crystal resonator **10** includes a quartz crystal element **11** having a shape of a thin chip, a first excitation electrode **14a** and a second excitation electrode **14b** that constitute a pair of excitation electrodes, a first extended electrode **15a** and a second extended electrode **15b** that constitute a pair of extended electrodes, and a first connection electrode **16a** and a second connection electrode **16b** that constitute a pair of connection electrodes.

(30) The quartz crystal element **11** has an upper surface **11A** and a lower surface **11B** facing away from each other. The upper surface **11A** is located on the side opposite the base member **30**, that is, the side facing a ceiling portion **41** of the lid member **40**, which will be described later. The lower surface **11B** is located on the side facing the base member **30**.

(31) The quartz crystal element **11** is, for example, an AT-cut quartz crystal element. The AT-cut quartz crystal element **11** is formed such that in an orthogonal coordinate system with an X-axis, a Y'-axis, and a Z'-axis intersecting one another, a plane parallel to a plane defined by the X-axis and the Z'-axis (hereinafter referred to as an "XZ' plane", the same applies to planes defined by other axes) is a principal plane, and a direction parallel to the Y'-axis is a thickness direction. For example, the AT-cut quartz crystal element **11** is formed by etching a quartz crystal substrate (e.g., a quartz crystal wafer) obtained by cutting and polishing a crystalline body of synthetic quartz crystal.

(32) The quartz crystal resonator **10** including the AT-cut quartz crystal element **11** has high frequency stability in a wide temperature range. In the AT-cut quartz crystal resonator **10**, a thickness shear vibration mode is used as a principal vibration. The rotation angles of the Y'-axis and the Z'-axis in the AT-cut quartz crystal element **11** may be inclined at an angle within the range from 35 degrees 15 minutes minus 5 degrees to 35 degrees 15 minutes plus 15 degrees. For the cut-angles of the quartz crystal element **11**, a cut different from the AT cut may be used. For example, BT cut, GT cut, SC cut, or other cuts may be used. The quartz crystal resonator may be a tuning fork quartz crystal resonator including a quartz crystal element having cut-angles of an element called a Z plate.

(33) The AT-cut quartz crystal element **11** has a plate shape with a long-side direction in which long sides parallel to the X-axis direction extend, a short-side direction in which short sides parallel to the Z'-axis direction extend, and a thickness direction in which a thickness parallel to the Y'-axis direction extends. The quartz crystal element **11** has a rectangular shape when the upper surface **11A** is viewed in plan, and has an excitation portion **17** that is centrally located and contributes to excitation and peripheral portions **18** and **19** adjacent to the excitation portion **17**. The excitation portion **17** and the peripheral portions **18** and **19** are each formed in a band shape across the width of the quartz crystal element **11** along the Z'-axis direction. The peripheral portion **18** is located on the -X side of the excitation portion **17**, and the peripheral portion **19** is located on the +X side of the excitation portion **17**.

(34) The planar shape of the quartz crystal element **11** when the upper surface **11A** is viewed in

plan is not limited to a rectangular shape. The planar shape of the quartz crystal element **11** may be a polygonal shape, a circular shape, an elliptical shape, or a combination thereof. The planar shape of the quartz crystal element **11** may be a tuning fork shape having a base portion and vibrating arm portions extending in parallel from the base portion. The quartz crystal element **11** may be provided with a slit for the purpose of suppressing vibration leakage and stress propagation. The shapes of the excitation portion **17** and the peripheral portions **18** and **19** of the quartz crystal element **11** are also not limited to the band shape across the width. For example, the planar shape of the excitation portion may be an island shape that is adjacent to peripheral portions also in the Z'-axis direction, and the planar shape of the peripheral portions may be shaped like a frame surrounding the excitation portion.

(35) The quartz crystal element **11** has a so-called mesa structure in which the thickness of the excitation portion **17** is larger than the thicknesses of the peripheral portions **18** and **19**. The quartz crystal element **11** having a mesa structure can suppress vibration leakage from the excitation portion **17**. The quartz crystal element **11** has a double-sided mesa structure. The excitation portion **17** protrudes beyond the peripheral portions **18** and **19** on both the upper surface **11A** side and the lower surface **11B** side. The boundary between the excitation portion **17** and the peripheral portion **18** and the boundary between the excitation portion **17** and the peripheral portion **19** have a tapered shape in which the thickness continuously changes, but may have a stepped shape in which the change in thickness is discontinuous. The boundaries may have a convex shape in which the amount of change in thickness continuously changes or a bevel shape in which the amount of change in thickness discontinuously changes. The quartz crystal element **11** may have a single-sided mesa structure in which the excitation portion **17** protrudes beyond the peripheral portions **18** and **19** on either the upper surface **11A** side or the lower surface **11B** side. The quartz crystal element **11** may have a so-called inverted mesa structure in which the thickness of the excitation portion **17** is smaller than the thicknesses of the peripheral portions **18** and **19**.

(36) The first excitation electrode **14a** and the second excitation electrode **14b** are provided on the excitation portion **17**. The first excitation electrode **14a** is provided on the upper surface **11A** side of the quartz crystal element **11**, and the second excitation electrode **14b** is provided on the lower surface **11B** side of the quartz crystal element **11**. In other words, the first excitation electrode **14a** is provided on the main surface on the lid member **40** side of the quartz crystal element **11**, and the second excitation electrode **14b** is provided on the main surface on the base member **30** side of the quartz crystal element **11**. The first excitation electrode **14a** and the second excitation electrode **14b** face away from each other with the quartz crystal element **11** interposed therebetween. When the upper surface **11A** of the quartz crystal element **11** is viewed in plan, the first excitation electrode **14a** and the second excitation electrode **14b** each have a rectangular shape and are disposed so as to substantially entirely overlap each other. The first excitation electrode **14a** and the second excitation electrode **14b** are each formed in a band shape across the width of the quartz crystal element **11** along the Z'-axis direction.

(37) The planar shape of the first excitation electrode **14a** and the second excitation electrode **14b** when the upper surface **11A** of the quartz crystal element **11** is viewed in plan is not limited to a rectangular shape. The planar shape of the first excitation electrode **14a** and the second excitation electrode **14b** may be a polygonal shape, a circular shape, an elliptical shape, or a combination thereof.

(38) The first extended electrode **15a** and the second extended electrode **15b** are provided on the peripheral portion **18**. The first extended electrode **15a** is provided on the upper surface **11A** side of the quartz crystal element **11**, and the second extended electrode **15b** is provided on the lower surface **11B** side of the quartz crystal element **11**. The first extended electrode **15a** electrically connects the first excitation electrode **14a** and the first connection electrode **16a** to each other. The second extended electrode **15b** electrically connects the second excitation electrode **14b** and the second connection electrode **16b** to each other. For example, as illustrated in FIG. 1, one end of the

first extended electrode **15a** is connected at the excitation portion **17** to the first excitation electrode **14a**, and the other end is connected at the peripheral portion **18** to the first connection electrode **16a** via a side electrode provided on a side surface connecting the upper surface **11A** and the lower surface **11B** of the quartz crystal element **11** to each other. One end of the second extended electrode **15b** is connected at the excitation portion **17** to the second excitation electrode **14b**, and the other end is connected at the peripheral portion **18** to the second connection electrode **16b** via the side electrode. To reduce stray capacitance, the first extended electrode **15a** and the second extended electrode **15b** are desirably separated from each other when the upper surface **11A** of the quartz crystal element **11** is viewed in plan. For example, the first extended electrode **15a** is provided in the +Z'-axis direction when viewed from the second extended electrode **15b**.

(39) The first connection electrode **16a** and the second connection electrode **16b** are each an electrode for electrically connecting the first excitation electrode **14a** and the second excitation electrode **14b** to the base member **30**, and are provided on the lower surface **11B** side of the quartz crystal element **11** at the peripheral portion **18**. The first connection electrode **16a** is provided at a corner portion formed of an end portion on the -X side and an end portion on the +Z' side of the quartz crystal element **11**, and the second connection electrode **16b** is provided at a corner portion formed of the end portion on the -X side and an end portion on the -Z' side of the quartz crystal element **11**.

(40) The first excitation electrode **14a**, the first extended electrode **15a**, and the first connection electrode **16a** are integrally formed, and the second excitation electrode **14b**, the second extended electrode **15b**, and the second connection electrode **16b** are integrally formed. The electrodes (the first excitation electrode **14a** and the second excitation electrode **14b**, the first extended electrode **15a** and the second extended electrode **15b**, and the first connection electrode **16a** and the second connection electrode **16b**) of the quartz crystal resonator **10** are each provided by, for example, laminating chromium (Cr) and gold (Au) in this order. Chromium is superior to gold in adhesion to the quartz crystal element **11**, and gold is superior to chromium in chemical stability. Thus, when the innermost surface in contact with the quartz crystal element **11** contains chromium and the outermost surface exposed to an internal space **49** contains gold, damage due to peeling off of the electrodes of the quartz crystal resonator **10** and variation in electrical conductivity due to oxidation of the electrodes can be reduced, and the quartz crystal resonator **10** with high reliability can be provided. The outermost surface of the electrodes of the quartz crystal resonator **10** desirably consists substantially of gold. The outermost surface of the first connection electrode **16a** corresponds to a contacting surface having an area in contact with a first electrically conductive holding member **36a**, and the outermost surface of the second connection electrode **16b** corresponds to a contacting surface having an area in contact with a second electrically conductive holding member **36b**. On a side surface connecting the innermost surface and the outermost surface of each electrode of the quartz crystal resonator **10** to each other, a layer containing chromium and a layer containing gold are exposed.

(41) Next, the base member **30** will be described.

(42) The base member **30** holds the quartz crystal resonator **10** such that the quartz crystal resonator **10** can be excited. The base member **30** includes a substrate **31** having an upper surface **31A** and a lower surface **31B** facing away from each other. The upper surface **31A** is located on the quartz crystal resonator **10** and the lid member **40** side and corresponds to a mounted surface on which the quartz crystal resonator **10** is mounted. The lower surface **31B** corresponds to a mounting surface to be joined to, for example, an external circuit board (not illustrated). The substrate **31** is, for example, a sintered material such as insulating ceramic (alumina). To suppress the generation of thermal stress, the substrate **31** is preferably made of a heat-resistant material. To reduce stress on the quartz crystal resonator **10** due to thermal hysteresis, the substrate **31** may be made of a material having a coefficient of thermal expansion close to that of the quartz crystal element **11**, such as quartz crystal.

(43) The base member **30** includes a first electrode pad **33a** and a second electrode pad **33b** that constitute a pair of electrode pads. The first electrode pad **33a** and the second electrode pad **33b** are provided on the upper surface **31A** of the substrate **31**. The first electrode pad **33a** and the second electrode pad **33b** are terminals for electrically connecting the quartz crystal resonator **10** to the base member **30**. To suppress a decrease in reliability due to oxidation, the outermost surface of each of the first electrode pad **33a** and the second electrode pad **33b** desirably contains gold, more desirably consists substantially of gold. For example, the first electrode pad **33a** and the second electrode pad **33b** may each have a two-layer structure including an underlying layer that improves adhesion to the substrate **31** and a surface layer that contains gold to suppress oxidation. The outermost surface of the first electrode pad **33a** corresponds to a contacting surface having an area in contact with the first electrically conductive holding member **36a**, and the outermost surface of the second electrode pad **33b** corresponds to a contacting surface having an area in contact with the second electrically conductive holding member **36b**.

(44) The base member **30** includes a first outer electrode **35a**, a second outer electrode **35b**, a third outer electrode **35c**, and a fourth outer electrode **35d**. The first outer electrode **35a** to the fourth outer electrode **35d** are provided on the lower surface **31B** of the substrate **31**. The first outer electrode **35a** and the second outer electrode **35b** are terminals for electrically connecting an external substrate (not illustrated) and the quartz crystal resonator unit **1** to each other. The third outer electrode **35c** and the fourth outer electrode **35d**, which are dummy electrodes to/from which an electrical signal or the like is not input/output, may be ground electrodes that ground the lid member **40** to improve the electromagnetic shielding function of the lid member **40**. The third outer electrode **35c** and the fourth outer electrode **35d** may be omitted.

(45) The first electrode pad **33a** and the second electrode pad **33b** are arranged side by side along the Z'-axis direction at an end portion in the -X-axis direction of the base member **30**. The first outer electrode **35a** and the second outer electrode **35b** are arranged side by side along the Z'-axis direction at the end portion in the -X-axis direction of the base member **30**. The third outer electrode **35c** and the fourth outer electrode **35d** are arranged side by side along the Z'-axis direction at an end portion in the +X-axis direction of the base member **30**. The first electrode pad **33a** is electrically connected to the first outer electrode **35a** via a first through electrode **34a** extending through the substrate **31** along the Y'-axis direction. The second electrode pad **33b** is electrically connected to the second outer electrode **35b** via a second through electrode **34b** extending through the substrate **31** along the Y'-axis direction.

(46) The first electrode pad **33a** and the second electrode pad **33b** may be electrically connected to the first outer electrode **35a** and the second outer electrode **35b**, respectively, via a side electrode provided on a side surface connecting the upper surface **31A** and the lower surface **31B** of the substrate **31** to each other. The first outer electrode **35a** to the fourth outer electrode **35d** may be castellated electrodes provided in a recessed manner in side surfaces of the substrate **31**.

(47) The base member **30** includes the first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** that constitute a pair of electrically conductive holding members. The first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** are for mounting the quartz crystal resonator **10** on the base member **30** and electrically connect the quartz crystal resonator **10** and the base member **30** to each other. The first electrically conductive holding member **36a** is joined to the first electrode pad **33a** and the first connection electrode **16a** and electrically connects the first electrode pad **33a** and the first connection electrode **16a** to each other. The second electrically conductive holding member **36b** is joined to the second electrode pad **33b** and the second connection electrode **16b** and electrically connects the second electrode pad **33b** and the second connection electrode **16b** to each other. The first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** hold the quartz crystal resonator **10** with a space left between the base member **30** and the quartz crystal resonator **10** so that the excitation portion **17** can be excited.

(48) The first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** are each a cured product of an electrically conductive adhesive including a thermosetting resin, a photocurable resin, or the like, and the main component of the first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** is a silicone resin. The first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** include electrically conductive particles, and the electrically conductive particles may be, for example, metal particles including silver (Ag). The first electrically conductive holding member **36a** bonds the first electrode pad **33a** and the first connection electrode **16a** to each other, and the second electrically conductive holding member **36b** bonds the second electrode pad **33b** and the second connection electrode **16b** to each other. The moduli of elasticity of silicone resins are more stable than those of epoxy resins in a wide temperature range. When the main component of the first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** is a silicone resin, the frequency-temperature characteristics of the quartz crystal resonator **10** are improved as compared to when the main component is an epoxy resin.

(49) The electrically conductive adhesive contains an epoxy compound having two or more glycidyl groups. In the present embodiment, the epoxy compound contained in the electrically conductive adhesive is 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane shown in FIG. 3. The content of 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane in the electrically conductive adhesive is, for example, 1 wt %. This improves the moisture resistance of the first electrically conductive holding member **36a** and the second electrically conductive holding member **36b**. Here, the improvement in moisture resistance means suppression of an increase in the contact resistance between the cured product of the electrically conductive adhesive and the adherends in a high-humidity environment.

(50) The epoxy compound contained in the electrically conductive adhesive is not limited to the above compound and may be any epoxy compound having two or more epoxy groups. However, to improve the moisture resistance, the epoxy compound having two or more glycidyl groups is desirable. The content of the epoxy compound in the electrically conductive adhesive is not limited to 1 wt %, but is desirably 0.1 wt % or more to improve the moisture resistance. The epoxy compound contained in the electrically conductive adhesive is desirably an organosilicon compound, more desirably an epoxy compound having a siloxane bond. For example, the epoxy compound may be an oligomer or a polymer having 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane as a structural unit. The component weight ratio of the electrically conductive adhesive is, for example, 60 wt % of electrically conductive particles, 20 wt % of a silicone resin composition, and 20 wt % of a solvent. The epoxy compound having two or more glycidyl groups is included in the silicone resin composition, and the content of the epoxy compound is the ratio of the weight of the epoxy compound to the total weight of the electrically conductive adhesive. That is, when the electrically conductive adhesive is composed of electrically conductive particles, a silicone resin composition, and a solvent, the content of the epoxy compound is the ratio of the weight of the epoxy compound to the sum of the weights of the electrically conductive particles, the silicone resin composition, and the solvent.

(51) The electrically conductive adhesive may contain additives in addition to silane or siloxane serving as a raw material for the silicone resin, the electrically conductive particles, and the epoxy compound having two or more epoxy groups. Examples of the additives include a tackifier, a filling agent, a thickener, a sensitizer, an antioxidant, and an antifoaming agent for the purpose of, for example, improving the workability and storage stability of the electrically conductive adhesive. In addition, a filler for the purpose of increasing the strength of the cured product or keeping the distance between the base member **30** and the quartz crystal resonator **10** may be added.

(52) The step of providing the first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** includes, for example, an application step of applying

an electrically conductive adhesive whose viscosity has been adjusted to be paste-like onto the first electrode pad **33a** and the second electrode pad **33b**; a mounting step of mounting the quartz crystal resonator **10** on the electrically conductive adhesive to let the electrically conductive adhesive that has been spread out on the surfaces of the first connection electrode **16a** and the second connection electrode **16b** support the quartz crystal resonator **10**; and a curing step of curing the electrically conductive adhesive. The step of providing the first electrically conductive holding member **36a** and the second electrically conductive holding member **36b** may include, before or after the mounting step, a temporary curing step of incompletely curing the electrically conductive adhesive. This can inhibit contact between the quartz crystal resonator **10** and the base member **30** due to collapse of the position of the quartz crystal resonator **10** before the curing step.

(53) Next, the lid member **40** will be described.

(54) The lid member **40** is joined to the base member **30** and forms, between the lid member **40** and the base member **30**, the internal space **49** in which the quartz crystal resonator **10** is accommodated. The lid member **40** may be made of any material, for example, an electrically conductive material such as metal. When the lid member **40** is made of an electrically conductive material, the lid member **40** is provided with an electromagnetic shielding function of reducing the entry and exit of electromagnetic waves to and from the internal space **49**.

(55) The lid member **40** includes a ceiling portion **41** having a flat shape and a side wall portion **42** connected to the edge of the ceiling portion **41** and extending in a direction perpendicular to the main surface of the ceiling portion **41**. The planar shape the ceiling portion **41** as viewed in plan from the direction normal to the main surface is, for example, a rectangular shape. The ceiling portion **41** faces the base member **30** with the quartz crystal resonator **10** interposed therebetween, and the side wall portion **42** surrounds the quartz crystal resonator **10** in a direction parallel to the XZ' plane. A tip portion of the side wall portion **42** extends in a frame-like shape such that the tip portion is closer to the base member **30** than the quartz crystal resonator **10** is.

(56) The lid member **40** may be made of a ceramic material, a semiconductor material, a resin material, or the like. The planar shape the ceiling portion **41** may be a polygonal shape, a circular shape, an elliptical shape, or a combination thereof.

(57) Next, the joining member **50** will be described.

(58) The joining member **50** is provided over the entire peripheries of the base member **30** and the lid member **40** and has a rectangular frame shape. When the upper surface **31A** of the base member **30** is viewed in plan, the first electrode pad **33a** and the second electrode pad **33b** are disposed inside the joining member **50**, and the joining member **50** is provided so as to surround the quartz crystal resonator **10**. The joining member **50** joins the tip portion of the side wall portion **42** of the lid member **40** and the upper surface **31A** of the substrate **31** of the base member **30** to each other to seal the internal space **49**. The joining member **50** desirably has high gas barrier properties, and more desirably has low moisture permeability. This joining member **50** is, for example, a cured product of an adhesive containing an epoxy resin as a main component. The resin-based adhesive constituting the joining member **50** may contain, for example, a vinyl compound, an acrylic compound, a urethane compound, or a silicone compound.

(59) The joining member **50** need not necessarily have a circumferentially continuous frame shape and may be provided discontinuously in the circumferential direction. The joining member **50** may be formed of, for example, a cured product of a silicon-based adhesive containing water glass and the like, a cured product of a calcium-based adhesive containing cement and the like, or Au—Sn alloy-based metal solder. When the joining member **50** is formed of metal solder, the base member **30** may be provided with a metallized layer for the purpose of improving the adhesion between the base member **30** and the joining member **50**. The joining member **50** may include a cured product of a resin-based adhesive and a coating that is less moisture-permeable than the cured product of a resin-based adhesive.

(60) (Moisture Resistance Evaluation)

(61) With reference to FIG. 4 to FIG. 7, a moisture resistance test to investigate the influence of humidity on the electrical conductivity between a metal electrode and a cured product of an electrically conductive adhesive will be described. FIG. 4 is a plan view schematically illustrating the structure of an evaluation substrate used in the moisture resistance test. FIG. 5 is a sectional view schematically illustrating the structure of the evaluation substrate used in the moisture resistance test. FIG. 6 is a graph schematically showing the evaluation results of an example. FIG. 7 is a graph schematically showing the evaluation results of a comparative example. In the graphs of FIG. 6 and FIG. 7, the horizontal axis represents the position of measured resistance, and the vertical axis represents resistance (Ω).

(62) As illustrated in FIG. 4, a first electrode pair E10, a second electrode pair E20, a third electrode pair E30, a fourth electrode pair E40, and a fifth electrode pair E50 are arranged in this order on an evaluation substrate B10. The first electrode pair E10 to the fifth electrode pair E50 are provided on an insulating substrate. The first electrode pair E10 includes a pair of measurement electrodes E11 and E12 and a bridge electrode E13 that connects the measurement electrode E11 and the measurement electrode E12 to each other. Likewise, the second electrode pair E20 includes a pair of measurement electrodes E21 and E22 and a bridge electrode E23, the third electrode pair E30 includes a pair of measurement electrodes E31 and E32 and a bridge electrode E33, the fourth electrode pair E40 includes a pair of measurement electrodes E41 and E42 and a bridge electrode E43, and the fifth electrode pair E50 includes a pair of measurement electrodes E51 and E52 and a bridge electrode E53. Central portions E1, E2, E3, E4, and E5 of the bridge electrodes E13 to E53 are covered by a cured electrically conductive adhesive E9. The cured electrically conductive adhesive E9 is substantially uniform in width and thickness and extends in a direction in which the first electrode pair E10 to the fifth electrode pair E50 are arranged. The first electrode pair E10 to the fifth electrode pair E50 are electrically connected together via the cured electrically conductive adhesive E9.

(63) The insulating substrate is an alumina substrate. The first electrode pair E10 to the fifth electrode pair E50 are metal electrodes having a laminated structure including an underlying layer formed of nickel and a surface layer formed of gold. The outermost surfaces of the first electrode pair E10 to the fifth electrode pair E50 are formed of gold. The cured electrically conductive adhesive E9 contains a silicone resin as a main component. The electrically conductive adhesive of the example contains 1% of 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane. The electrically conductive adhesive of the comparative example is the same as the electrically conductive adhesive of the example except that 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane is not contained.

(64) As illustrated in FIG. 5, changes of a resistance 1-2 between the central portion E1 and the central portion E2, a resistance 1-3 between the central portion E1 and the central portion E3, a resistance 1-4 between the central portion E1 and the central portion E4, and a resistance 1-5 between the central portion E1 and the central portion E5 were measured, whereby the moisture resistance of the cured electrically conductive adhesive E9 was evaluated. The resistance 1-2 is determined by the contact resistance between the central portion E1 and the cured electrically conductive adhesive E9, the resistance of the cured electrically conductive adhesive E9 between the central portion E1 and the central portion E2, and the contact resistance between the cured electrically conductive adhesive E9 and the central portion E2. Similarly, the resistance 1-3, the resistance 1-4, and the resistance 1-5 are each determined by the resistance of the cured electrically conductive adhesive E9 itself and the contact resistance between the cured electrically conductive adhesive E9 and the central portion of each electrode pair.

(65) To offset the influence of factors other than the central portions E1 and E2 and the cured electrically conductive adhesive E9, the resistance 1-2 was measured as follows.

(66) The resistance between the measurement electrode E11 and the measurement electrode E21, the resistance between the measurement electrode E11 and the measurement electrode E22, the

resistance between the measurement electrode E12 and the measurement electrode E22, and the resistance between the measurement electrode E12 and the measurement electrode E21 were measured and summed up. This calculation result was taken as A.

(67) The resistance between the measurement electrode E11 and the measurement electrode E12 and the resistance between the measurement electrode E21 and the measurement electrode E22 were measured, summed up, and doubled. This calculation result was taken as B.

(68) B was subtracted from A, and the result was divided by 4 to calculate the resistance 1-2.

(69) The resistance 1-3, the resistance 1-4, and the resistance 1-5 were calculated in the same manner.

(70) As shown in FIG. 6, in the example, the measurement was performed in an initial state, a state after being left to stand for 100 hours in an environment of a temperature of 85° C. and a humidity of 85% RH (hereinafter referred to as “85° C. and 85% RH”) in a constant temperature and humidity chamber, and a state after being left to stand for 500 hours in 85° C. and 85% RH. In FIG. 6, a line graph indicating the resistance in the initial state is represented as “INITIAL”, a line graph indicating the resistance in the state after being left to stand for 100 hours as “AFTER 100 h”, and a line graph indicating the resistance in the state after being left to stand for 500 hours as “AFTER 500 h”.

(71) Even in the state after being left to stand for 500 hours, the resistance 1-2, the resistance 1-3, the resistance 1-4, and the resistance 1-5 were all 1Ω or less, deterioration of the resistance from the initial state was hardly observed.

(72) As shown in FIG. 7, in the comparative example, the measurement of resistance was performed in an initial state, a state after being left to stand for 150 hours in 85° C. and 85% RH, and a state after being left to stand for 530 hours in 85° C. and 85% RH. In FIG. 7, a line graph indicating the resistance in the initial state is represented as “INITIAL”, a line graph indicating the resistance in the state after being left to stand for 150 hours as “AFTER 150 h”, and a line graph indicating the resistance in the state after being left to stand for 530 hours as “AFTER 530 h”.

(73) In the state after being left to stand for 530 hours, the resistance 1-3, the resistance 1-4, and the resistance 1-5 were more than 1Ω, that is, deterioration from the initial state was observed. In particular, the resistance 1-4 was more than 4Ω in the state after being left to stand for 530 hours. Since the resistance 1-3 and the resistance 1-5 are higher than the resistance 1-4 in the state after being left to stand for 530 hours, the major factor of the increase in resistance is probably not an increase in the resistance of the cured electrically conductive adhesive itself but an increase in the contact resistance between the cured electrically conductive adhesive and the central portion of each electrode pair.

(74) The evaluation results of the example and the comparative example have shown that the presence of 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane in a resin composition of an electrically conductive adhesive can suppress an increase in the contact resistance between a metal electrode and a cured product of the electrically conductive adhesive containing a silicone resin as a main component in a high-humidity environment.

(75) Next, with reference to FIG. 9 to FIG. 14, the relationship between the content of 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane and the moisture resistance of cured electrically conductive adhesives will be described. FIG. 9 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 0 wt %. FIG. 10 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 0.1 wt %. FIG. 11 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 0.5 wt %. FIG. 12 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 1 wt %. FIG. 13 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 5 wt %. FIG. 14 is a graph showing the results of an evaluation of moisture resistance in the case of an epoxy compound content of 10 wt %.

(76) In the moisture resistance test, changes in resistance in 85° C. and 85% RH were measured and compared in the same manner as the method of evaluation of the example and the comparative example shown in FIG. 4 to FIG. 7. In FIG. 9 to FIG. 14, “INITIAL” represents resistance in an initial state before being put into 85° C. and 85% RH, “AFTER 100 h” represents resistance at 100 hours after being put into 85° C. and 85% RH, “AFTER 500 h” represents resistance at 500 hours after being put into 85° C. and 85% RH, and “AFTER 1000 h” represents resistance at 1000 hours after being put into 85° C. and 85% RH. FIG. 9 corresponds to the evaluation results of the comparative example, and FIG. 12 corresponds to the evaluation results of the example. Regardless of the content of 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane, the resistance in the initial state was substantially 0Ω. Therefore, the values of resistance at 100, 500, and 1000 hours each correspond to the amount of resistance change from the initial state. Larger amounts of resistance change indicate lower moisture resistance.

(77) When the content of 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane was 0 wt %, or when the epoxy compound was not contained, the resistance had already increased at 100 hours, had increased to about 5.0Ω at maximum at 500 hours, and had exceeded 15.0Ω at maximum at 1000 hours. When the content of 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane was 0.1 wt %, the resistance showed almost no increase until at 100 hours, had started to increase at 500 hours, and had increased to about 4.0Ω at maximum at 1000 hours. When the content of 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane was 0.5 to 10 wt %, the resistance showed almost no increase even at 1000 hours.

(78) The above shows that the moisture resistance of a cured product of a silicone electrically conductive adhesive is improved by adding 0.1 wt % or more of an epoxy compound having two or more epoxy groups, and is further improved by adding 0.5 wt % or more of the epoxy compound. However, when the epoxy compound is added to the silicone electrically conductive adhesive, the viscosity decreases. In particular, when the content of the epoxy compound is more than 10 wt %, the decrease in viscosity may reduce workability, thus resulting in an increased rate of occurrence of defective quartz crystal resonator units. Therefore, to improve the moisture resistance while maintaining the workability, the content of the epoxy compound is desirably 0.1 wt % to 10 wt %, more desirably 0.5 wt % to 10 wt %. When the content of the epoxy compound is more than 10 wt %, the characteristics of the silicone resin may be lost from the cured product of the electrically conductive adhesive. For example, when the content of the epoxy compound is more than 10 wt %, the modulus of elasticity of the cured product of the electrically conductive adhesive may greatly vary with temperature changes, thus degrading the frequency-temperature characteristics of the quartz crystal resonator 10. Therefore, to inhibit the degradation of the frequency-temperature characteristics, the content of the epoxy compound is desirably 10 wt % or less.

(79) Next, the moisture resistance of cured products of electrically conductive adhesives containing epoxy compounds other than 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane will be described with reference to FIG. 15 to FIG. 22. FIG. 15 shows a structural formula of an epoxy compound added in a first modified example. FIG. 16 is a graph showing the results of an evaluation of moisture resistance in the first modified example. FIG. 17 shows a structural formula of an epoxy compound added in a second modified example. FIG. 18 is a graph showing the results of an evaluation of moisture resistance in the second modified example. FIG. 19 shows a structural formula of an epoxy compound added in a third modified example. FIG. 20 is a graph showing the results of an evaluation of moisture resistance in the third modified example. FIG. 21 shows a structural formula of an epoxy compound added in a fourth modified example. FIG. 22 is a graph showing the results of an evaluation of moisture resistance in the fourth modified example.

(80) In the moisture resistance test for the first to fourth modified examples, changes in resistance in 85° C. and 85% RH were measured and compared in the same manner as the method of evaluation of the example and the comparative example shown in FIG. 4 to FIG. 7. “INITIAL” represents resistance in an initial state before being put into 85° C. and 85% RH, “AFTER 100 h”

represents resistance at 100 hours after being put into 85° C. and 85% RH, “AFTER 240 h” represents resistance at 240 hours after being put into 85° C. and 85% RH, “AFTER 500 h” represents resistance at 500 hours after being put into 85° C. and 85% RH, and “AFTER 1000 h” represents resistance at 1000 hours after being put into 85° C. and 85% RH. The resistance in the initial state in each of the first to fourth modified examples was substantially 0 Ω . Therefore, the values of resistance at 100, 240, 500, and 1000 hours each correspond to the amount of resistance change from the initial state. Larger amounts of resistance change indicate lower moisture resistance.

(81) The first to fourth modified examples have the same configuration as that of the present embodiment except that the type of epoxy compound contained in the electrically conductive adhesive is different. The epoxy compound contained in the electrically conductive adhesive of the first modified example is 1,4-butanediol diglycidyl ether shown in FIG. 15. The epoxy compound contained in the electrically conductive adhesive of the second modified example is neopentylglycol diglycidyl ether shown in FIG. 17. The epoxy compound contained in the electrically conductive adhesive of the third modified example is trimethylolpropane triglycidyl ether shown in FIG. 19. The epoxy compound contained in the electrically conductive adhesive of the fourth modified example is polyethylene glycol diglycidyl ether shown in FIG. 21. The epoxy compound content in each of the first to fourth modified examples is 1.0 wt %.

(82) In the case of the first modified example shown in FIG. 16, the resistance showed almost no increase until at 100 hours, had started to increase at 240 hours, and had increased to about 14 Ω at maximum at 1000 hours. In the case of the second modified example shown in FIG. 18, the resistance showed almost no increase until at 240 hours, had started to increase at 500 hours, and had increased to about 5 to 6 Ω at maximum at 1000 hours. In the case of the third modified example shown in FIG. 20, the resistance showed almost no increase until at 240 hours, had started to increase at 500 hours, and had increased to about 5 to 6 Ω at maximum at 1000 hours. In the case of the fourth modified example shown in FIG. 22, the resistance showed almost no increase even at 1000 hours.

(83) In the case of the comparative example shown in FIG. 9, the resistance had started to increase at 100 hours and had exceeded 15 Ω at maximum at 1000 hours. In the case of the example shown in FIG. 12, the resistance had remained substantially 0 Ω at 1000 hours. That is to say, the start of the increase in resistance in the first to fourth modified examples is slower than in the comparative example. The maximum value of resistance in the first modified example is larger than the maximum value of resistance in the example and smaller than the maximum value of resistance in the comparative example. The maximum values of resistance in the second and third modified examples are larger than the maximum value of resistance in the example and less than half of the maximum value of resistance in the comparative example. The maximum value of resistance in the fourth modified example is comparable to the maximum value of resistance in the example. In other words, the moisture resistance in the first to fourth modified examples, as in the case of the example, is higher than that in the comparative example. The moisture resistance in the first modified example is higher than the moisture resistance in the comparative example. The moisture resistance in the second modified example is substantially comparable to the moisture resistance in the third modified example and higher than the moisture resistance in the first modified example. The moisture resistance in the fourth modified example is substantially comparable to the moisture resistance in the example and higher than the moisture resistance in the second and third modified examples.

(84) The above shows that the epoxy compound that improves the moisture resistance of a cured product of a silicone electrically conductive adhesive is not limited to 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane and may be any epoxy compound having two or more epoxy groups. In addition, from the fact that the moisture resistance in the example and the fourth modified example is particularly high, it is presumed that the epoxy compound that

improves the moisture resistance has features such as being an organosilicon compound that has a siloxane bond and having two epoxy groups that are linked through a long main chain (e.g., nine or more atoms).

(85) As described above, in the quartz crystal resonator unit **1** according to the present embodiment, the electrically conductive holding member **36a** that connects the connection electrode **16a** and the electrode pad **33a** to each other and the electrically conductive holding member **36b** that connects the connection electrode **16b** and the electrode pad **33b** to each other are each a cured product of an electrically conductive adhesive containing a silicone resin as a main component, and the electrically conductive adhesive contains an epoxy compound having two or more epoxy groups.

(86) With this configuration, deterioration of the resistance between a base member and a quartz crystal resonator in a high-humidity environment can be reduced while improving the frequency-temperature characteristics of the quartz crystal resonator. Specifically, increases in the contact resistance between electrically conductive holding members and connection electrodes and the contact resistance between the electrically conductive holding members and electrode pads can be suppressed.

(87) In the quartz crystal resonator unit **1** according to the present embodiment, the outermost surfaces of the electrode pads **33a** and **33b** contain gold.

(88) If the electrically conductive adhesive contains an epoxy compound having two or more epoxy groups, an increase in the contact resistance between the electrically conductive adhesive and the electrode pads in a high-humidity environment can be suppressed even if gold, which has low adhesion to resin-based adhesives, is contained in the outermost surfaces of the electrode pads. This eliminates the need for surface treatment (e.g., Ag sputtering) on the electrode pads for improving the adhesion to the electrically conductive adhesive, thus preventing a decrease in corrosion resistance of the electrode pads due to the surface treatment. Consequently, a quartz crystal resonator unit having high corrosion resistance and high moisture resistance can be provided.

(89) In the quartz crystal resonator unit **1** according to the present embodiment, the outermost surfaces of the connection electrodes **16a** and **16b** contain gold.

(90) If the electrically conductive adhesive contains an epoxy compound having two or more epoxy groups, an increase in the contact resistance between the electrically conductive adhesive and the connection electrodes in a high-humidity environment can be suppressed even if gold, which has low adhesion to resin-based adhesives, is contained in the outermost surfaces of the connection electrodes. This eliminates the need for surface treatment (e.g., Ag sputtering) on the connection electrodes for improving the adhesion to the electrically conductive adhesive, thus preventing a decrease in corrosion resistance of the connection electrodes due to the surface treatment.

Consequently, a quartz crystal resonator unit having high corrosion resistance and high moisture resistance can be provided.

(91) In the quartz crystal resonator unit **1** according to the present embodiment, the joining member **50** that joins the base member **30** and the lid member **40** to each other to seal the internal space **49** is further included, and the joining member **50** is a cured product of a resin-based adhesive.

(92) When the base member and the lid member are joined with a resin-based adhesive, the quartz crystal resonator unit can be produced at lower cost but has higher moisture permeability than when they are joined with a metal. Even in such a case, deterioration of the resistance between the base member and the quartz crystal resonator in a high-humidity environment can be reduced. Consequently, a quartz crystal resonator unit having high moisture resistance can be provided while reducing the production cost.

(93) In the quartz crystal resonator unit **1** according to the present embodiment, the epoxy compound contained in the electrically conductive adhesive is 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane, which is an organosilicon compound having a siloxane bond.

(94) In the quartz crystal resonator unit **1** according to the present embodiment, the content of the

epoxy compound in the electrically conductive adhesive is preferably 0.5 wt % to 10.0 wt %.

(95) With this configuration, an increase in the resistance of the electrically conductive holding members **36a** and **36b** in a high-humidity environment can be suppressed while suppressing a decrease in the frequency and temperature dependence of the quartz crystal resonator unit **1**.

(96) Embodiments according to the present invention are applicable not only to quartz crystal resonator units but also to piezoelectric resonator units. An example of the piezoelectric resonator units is a quartz crystal resonator unit including a quartz crystal resonator. The quartz crystal resonator uses a quartz crystal element as a piezoelectric element that is excited by the piezoelectric effect. The piezoelectric element may be formed of any piezoelectric material such as a piezoelectric single crystal, a piezoelectric ceramic, a piezoelectric thin film, or a piezoelectric polymer film. One example of the piezoelectric single crystal is lithium niobate (LiNbO_3). Similarly, examples of the piezoelectric ceramic include barium titanate (BaTiO_3), lead titanate (PbTiO_3), lead zirconate titanate ($\text{Pb}(\text{Zr}_x\text{Ti}_{1-x})\text{O}_3$; PZT), aluminum nitride (AlN), lithium niobate (LiNbO_3), lithium metaniobate (LiNb_2O_6), bismuth titanate ($\text{Bi}_4\text{Ti}_3\text{O}_{12}$), lithium tantalate (LiTaO_3), lithium tetraborate ($\text{Li}_2\text{B}_4\text{O}_7$), langasite ($\text{La}_3\text{Ga}_5\text{SiO}_{14}$), and tantalum pentoxide (Ta_2O_5). The piezoelectric thin film may be formed by depositing the piezoelectric ceramic on a substrate made of quartz, sapphire, or the like by, for example, sputtering. Examples of the piezoelectric polymer film include polylactic acid (PLA), polyvinylidene fluoride (PVDF), and vinylidene fluoride/trifluoroethylene (VDF/TrFE) copolymers. The piezoelectric materials given above may be stacked on top of each other or may be stacked on another member.

(97) Embodiments according to the present invention are applicable not only to piezoelectric resonator units but also to any electronic component that includes a first adherend portion having electrical conductivity, a second adherend portion having electrical conductivity, and an electrically conductive adhesive portion that connects the first adherend portion and the second adherend portion to each other, wherein the electrically conductive adhesive portion is a cured product of an electrically conductive adhesive containing a silicone resin as a main component, and the electrically conductive adhesive contains an epoxy compound having two or more epoxy groups. For example, in the case of the quartz crystal resonator unit **1**, the first adherend portion corresponds to the connection electrodes **16a** and **16b**, the second adherend portion corresponds to the electrode pads **33a** and **33b**, the electrically conductive adhesive portion corresponds to the electrically conductive holding members **36a** and **36b**, and the electronic component corresponds to the quartz crystal resonator unit **1**. In other words, any electronic component containing bonded portions that are bonded to each other with an electrically conductive adhesive portion interposed therebetween and are required to be electrically connected to the electrically conductive adhesive portion can be an embodiment of the present invention without particular limitation. Examples of such electronic components include resonator units other than quartz crystal resonator units, transistors, diodes, capacitors, inductors, resistors, integrated circuits (ICs), image pickup devices, display devices, sensors, and MEMS devices.

(98) For example, when the electronic component is a resonator having a mechanical vibration portion, the electrically conductive adhesive portion is subjected to a vibration load. Thus, when the adhesion between the adherend portions and the electrically conductive adhesive portion is insufficient, the contact resistance at the interfaces between the adherend portions and the electrically conductive adhesive portion increases due to the low adhesion, and in the worst case, separation occurs, thus resulting in loss of the electrical conductivity. Even in the case of such an electronic component having a configuration in which the interfaces between adherend portions and an electrically conductive adhesive portion are subjected to a load due to an impact during transportation, displacement during operation, or the like, an increase in contact resistance in a high-humidity environment can be suppressed according to an embodiment of the present invention.

(99) Hereinafter, the configuration of an electronic device according to another embodiment of the present invention will be described. The following embodiment will be described only by referring to points different from the first embodiment while omitting descriptions of points common to the first embodiment. In particular, similar advantages obtained by similar configurations will not be repeated.

Second Embodiment

(100) The configuration of an electronic device **100** according to a second embodiment will be described with reference to FIG. **8**. FIG. **8** is a sectional view schematically illustrating the configuration of the electronic device according to the second embodiment.

(101) The electronic device **100** according to the second embodiment includes an electronic component **70**, a substrate **90** including a metal layer **91**, and an electrically conductive adhesive portion **80** that connects the electronic component **70** and the metal layer **91** of the substrate **90** to each other. The electrically conductive adhesive portion **80** is a cured product of an electrically conductive adhesive containing a silicone resin as a main component, and the electrically conductive adhesive contains an epoxy compound having two or more epoxy groups. In FIG. **8**, for example, a quartz crystal resonator unit is depicted as the electronic component **70**. However, the electronic component **70** is not limited to the quartz crystal resonator unit and may be, for example, a resonator unit other than the quartz crystal resonator unit, a transistor, a diode, a capacitor, an inductor, a resistor, an integrated circuit (IC), an image pickup device, a display device, a sensor, a MEMS device.

(102) The electrically conductive adhesive portion that is a cured product of an electrically conductive adhesive containing a silicone resin as a main component, the electrically conductive adhesive containing an epoxy compound having two or more epoxy groups, may be used for electrical connection outside the electronic component as in the second embodiment, or may be used for electrical connection inside the electronic component as in the first embodiment.

(103) Hereinafter, some or all of the embodiments of the present invention will be additionally described, and the advantages thereof will be described. The present invention is not limited to the following additional descriptions.

(104) One aspect of the present invention provides a quartz crystal resonator unit including: a quartz crystal resonator including an excitation electrode and a connection electrode electrically connected to the excitation electrode; a base member including an electrode pad; an electrically conductive holding member that connects the connection electrode and the electrode pad to each other; and a lid member attached to the base member so as to form an internal space between the lid member and the base member in which the quartz crystal resonator is accommodated. The electrically conductive holding member is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main component and (2) an epoxy compound having two or more epoxy groups.

(105) With this configuration, deterioration of the resistance between the base member and the quartz crystal resonator in a high-humidity environment can be reduced while improving the frequency-temperature characteristics of the quartz crystal resonator. Specifically, increases in the contact resistance between the electrically conductive holding member and the connection electrode and the contact resistance between the electrically conductive holding member and the electrode pad can be suppressed. That is, a quartz crystal resonator unit having high moisture resistance can be provided.

(106) In one aspect, the epoxy groups are each a part of a glycidyl group.

(107) If the epoxy groups of the epoxy compound contained in the electrically conductive adhesive are each a part of a glycidyl group, a quartz crystal resonator unit having particularly high moisture resistance can be provided.

(108) In one aspect, the outermost surface of the electrode pad contains gold.

(109) If the electrically conductive adhesive contains an epoxy compound having two or more

epoxy groups, an increase in the contact resistance between the electrically conductive adhesive and the electrode pad in a high-humidity environment can be suppressed even if gold, which has low adhesion to resin-based adhesives, is contained in the outermost surface of the electrode pad. This eliminates the need for surface treatment (e.g., Ag sputtering) on the electrode pad for improving the adhesion to the electrically conductive adhesive, thus preventing a decrease in corrosion resistance of the electrode pad due to the surface treatment. Consequently, a quartz crystal resonator unit having high corrosion resistance and high moisture resistance can be provided.

(110) In one aspect, the outermost surface of the connection electrode contains gold.

(111) If the electrically conductive adhesive contains an epoxy compound having two or more epoxy groups, an increase in the contact resistance between the electrically conductive adhesive and the connection electrode in a high-humidity environment can be suppressed even if gold, which has low adhesion to resin-based adhesives, is contained in the outermost surface of the connection electrode. This eliminates the need for surface treatment (e.g., Ag sputtering) on the connection electrode for improving the adhesion to the electrically conductive adhesive, thus preventing a decrease in corrosion resistance of the connection electrode due to the surface treatment. Consequently, a quartz crystal resonator unit having high corrosion resistance and high moisture resistance can be provided.

(112) In one aspect, a joining member that joins the base member and the lid member to each other to seal the internal space is further included, and the joining member is a resin-based adhesive.

(113) When the base member and the lid member are joined with a resin-based adhesive, the quartz crystal resonator unit can be produced at lower cost but has higher moisture permeability than when they are joined with a metal. Even in such a case, deterioration of the resistance between the base member and the quartz crystal resonator in a high-humidity environment can be reduced. Consequently, a quartz crystal resonator unit having high moisture resistance can be provided while reducing the production cost.

(114) In one aspect, the epoxy compound is an organosilicon compound.

(115) In one aspect, the epoxy compound has a siloxane bond.

(116) In one aspect, the epoxy compound is 1,3-bis(3-glycidoxypyl)-1,1,3,3-tetramethyldisiloxane or includes 1,3-bis(3-glycidoxypyl)-1,1,3,3-tetramethyldisiloxane as a structural unit.

(117) In the quartz crystal resonator unit **1** according to the present embodiment, a content of the epoxy compound in the electrically conductive adhesive is 0.5 wt % to 10.0 wt %.

(118) With this configuration, an increase in the resistance of the electrically conductive holding members in a high-humidity environment can be suppressed while suppressing a decrease in the frequency and temperature dependence of the quartz crystal resonator unit.

(119) Another aspect of the present invention provides an electronic device including an electronic component, a substrate including a metal layer, and an electrically conductive adhesive portion that connects the electronic component and the metal layer of the substrate to each other. The electrically conductive adhesive portion is a cured product of an electrically conductive adhesive containing a silicone resin as a main component, and the electrically conductive adhesive contains an epoxy compound having two or more epoxy groups.

(120) With this configuration, deterioration of the resistance between the electronic component and the substrate in a high-humidity environment can be reduced. That is, an electronic device having high moisture resistance can be provided.

(121) Another aspect of the present invention provides an electronic component including a first adherend portion having electrical conductivity, a second adherend portion having electrical conductivity, and an electrically conductive adhesive portion that connects the first adherend portion and the second adherend portion to each other. The electrically conductive adhesive portion is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main

component and (2) an epoxy compound having two or more epoxy groups.

(122) With this configuration, deterioration of the resistance between the first adherend portion and the second adherend portion in a high-humidity environment can be reduced. That is, an electronic device having high moisture resistance can be provided.

(123) In one aspect, the first adherend portion is a connection electrode provided in or on a resonator having a mechanical vibration portion.

(124) Even in the case of an electronic component, such as an electronic element, having a configuration in which the interfaces between adherend portions and an electrically conductive adhesive portion are subjected to a load due to an impact during transportation, displacement during operation, or the like, an increase in contact resistance in a high-humidity environment can be suppressed according to an embodiment of the present invention.

(125) As described above, the aspects of the present invention can provide a quartz crystal resonator unit with improved reliability, an electronic component, and an electronic device.

(126) The embodiments described above are presented for a better understanding of the present invention and should not be construed as limiting the present invention. The present invention can be modified or improved without departing from the gist of the invention, and equivalents to the present invention are also included in the present invention. That is, appropriate design modifications made to the embodiments by those skilled in the art are also within the scope of the present invention as long as the modifications have the features of the present invention. For example, the elements included in the embodiments and the arrangements, materials, conditions, shapes, sizes, and the like of the elements are not limited to those exemplified in the embodiments and can be appropriately changed. For example, the resonator and the resonator unit of the present invention can be used as a timing device or a load sensor. The elements included in the embodiments can be combined with each other as long as it is technically possible, and such combinations are also within the scope of the present invention as long as the combinations have the features of the present invention.

REFERENCE SIGNS LIST

(127) **1** quartz crystal resonator unit **10** quartz crystal resonator **11** quartz crystal element **14a**, **14b** excitation electrode **15a**, **15b** extended electrode **16a**, **16b** connection electrode **30** base member **31** substrate **33a**, **33b** electrode pad **34a**, **34b** through electrode **35a** to **35d** outer electrode **36a**, **36b** electrically conductive holding member **40** lid member **50** joining member

Claims

1. A quartz crystal resonator unit comprising: a quartz crystal resonator including an excitation electrode and a connection electrode electrically connected to the excitation electrode; a base member including an electrode pad; an electrically conductive holding member that connects the connection electrode and the electrode pad to each other; and a lid member attached to the base member so as to form an internal space between the lid member and the base member in which the quartz crystal resonator is accommodated, wherein the electrically conductive holding member is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main component and (2) an epoxy compound having two or more epoxy groups.
2. The quartz crystal resonator unit according to claim 1, wherein the two or more epoxy groups are each a part of a glycidyl group.
3. The quartz crystal resonator unit according to claim 1, wherein an outermost surface of the electrode pad contains gold.
4. The quartz crystal resonator unit according to claim 1, wherein an outermost surface of the connection electrode contains gold.
5. The quartz crystal resonator unit according to claim 1, further comprising a joining member that joins the base member and the lid member to each other to seal the internal space, wherein the

joining member is a cured product of a resin-based adhesive.

6. The quartz crystal resonator unit according to claim 1, wherein the epoxy compound is an organosilicon compound.

7. The quartz crystal resonator unit according to claim 1, wherein the epoxy compound has a siloxane bond.

8. The quartz crystal resonator unit according to claim 1, wherein the epoxy compound is 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane or includes 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane as a structural unit.

9. The quartz crystal resonator unit according to claim 1, wherein the electrically conductive adhesive contains 0.5 wt % to 10.0 wt % of the epoxy compound.

10. The quartz crystal resonator unit according to claim 1, wherein the electrically conductive adhesive contains 0.1 wt % to 10.0 wt % of the epoxy compound.

11. An electronic device comprising: an electronic component; a substrate including a metal layer; and an electrically conductive adhesive portion that connects the electronic component and the metal layer of the substrate to each other, wherein the electrically conductive adhesive portion is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main component and (2) an epoxy compound having two or more epoxy groups.

12. The electronic device according to claim 11, wherein the two or more epoxy groups are each a part of a glycidyl group.

13. The electronic device according to claim 11, wherein the epoxy compound is an organosilicon compound.

14. The electronic device according to claim 11, wherein the epoxy compound has a siloxane bond.

15. The electronic device according to claim 11, wherein the epoxy compound is 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane or includes 1,3-bis(3-glycidoxypropyl)-1,1,3,3-tetramethyldisiloxane as a structural unit.

16. The electronic device according to claim 11, wherein the electrically conductive adhesive contains 0.5 wt % to 10.0 wt % of the epoxy compound.

17. An electronic component comprising: a first adherend portion having electrical conductivity; a second adherend portion having electrical conductivity; and an electrically conductive adhesive portion that connects the first adherend portion and the second adherend portion to each other, wherein the electrically conductive adhesive portion is a cured product of an electrically conductive adhesive containing (1) a silicone resin as a main component and (2) an epoxy compound having two or more epoxy groups.

18. The electronic component according to claim 17, wherein the first adherend portion is a connection electrode provided in or on a resonator having a mechanical vibration portion.

19. The electronic component according to claim 17, wherein the two or more epoxy groups are each a part of a glycidyl group.

20. The electronic component according to claim 17, wherein the electrically conductive adhesive contains 0.5 wt % to 10.0 wt % of the epoxy compound.
